

Introduction

17-313 Fall 2026

Foundations of Software Engineering

<https://cmu-17313q.github.io>

Eduardo Feo Flushing

Introductions

Eduardo Feo Flushing



B.Sc. In Computer Eng., Universidad Simón Bolívar, Venezuela - 2007



M.Sc. in Informatics, University of Trento, Italy - 2010



M.Sc. in Software Systems Engineering, RWTH-Aachen, Germany- 2010



Ph.D. in Informatics, University of Lugano - IDSIA, Switzerland - 2017

Postdoc, CMU Qatar - Fall 2018



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SOFTWARE IS EATING THE WORLD

Software is everywhere

**(Bad)Software is
everywhere**



Ariane 5, 1996: self-destructed 37 seconds after launch due to a reused Ariane 4 software module

THE VERGE

Figure 1,
Relative

BOEING

Probabilistic
Consequence Gra

121
121
135
92
121
92



REDLINE

The many human
errors that brought down
the Boeing 737 Max

Catastrophic

11.2 IN
(12.55 M)

9 IN (5.72 M)

Aut
Effects
Occupants

Security

CrowdStrike accepts award for 'most epic fail' after global IT outage

Anthony Ha / 10:40 AM PDT • August 11, 2024

[Comment](#)



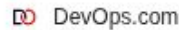
Image Credits: Lorenzo Franceschi-Bicchlerai / TechCrunch



Claude AI Recovering After Widespread Outage on Wednesday

Gael Cooper ... Anthropic's Claude is coming back from a major outage on Wednesday. The site experienced major errors for part of the day, but...

3 weeks ago



GitHub Hit by Widespread Outage, Halting Work for Global Developers

A major technical outage struck GitHub on Monday morning, disrupting workflows for millions of developers worldwide for several hours as...

4 days ago



ChatGPT Down: OpenAI Confirms Login And Signup Outage — Latest Updates

OpenAI has confirmed that logins and signups on chatgpt.com are failing, with users seeing an authentication key error.

2 days ago



Failure is not always a coding problem

Vasa



Vasa



Image credit: perdurabo10.tripod.com/ships/id170.html

What happened is now called “Vasa syndrome”

- Changing shipbuilding orders
- No specifications for modified keel
- Shifting armaments requirements

Requirements

- Shipwright's death

Teams

- No way to calculate stability, stiffness, or sailing characteristics

Metrics

- Failed pre-launch stability tests

QA

What makes creating software hard?

Think for 30 seconds. Then share one idea with someone next to you.

No Silver Bullet

Accidental Complexity

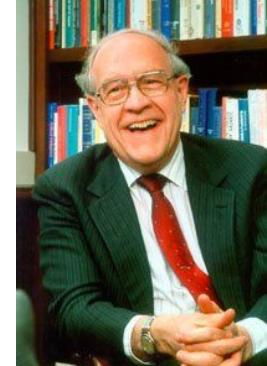
Complexity caused by the tools we use

- Programming language limitations
- Coding (Syntax, bugs)
- Deployment
- Compatibility issues
- Manual testing that could be automated

Essential Complexity

Complexity inherent in the problem

- Understanding what the system must do
- Modeling a complex domain
- Managing conflicting requirements and trade-offs
- Supporting many constraints and use cases



No Silver Bullet: Essence and Accidents of Software Engineering

by Frederick P. Brooks, Jr.

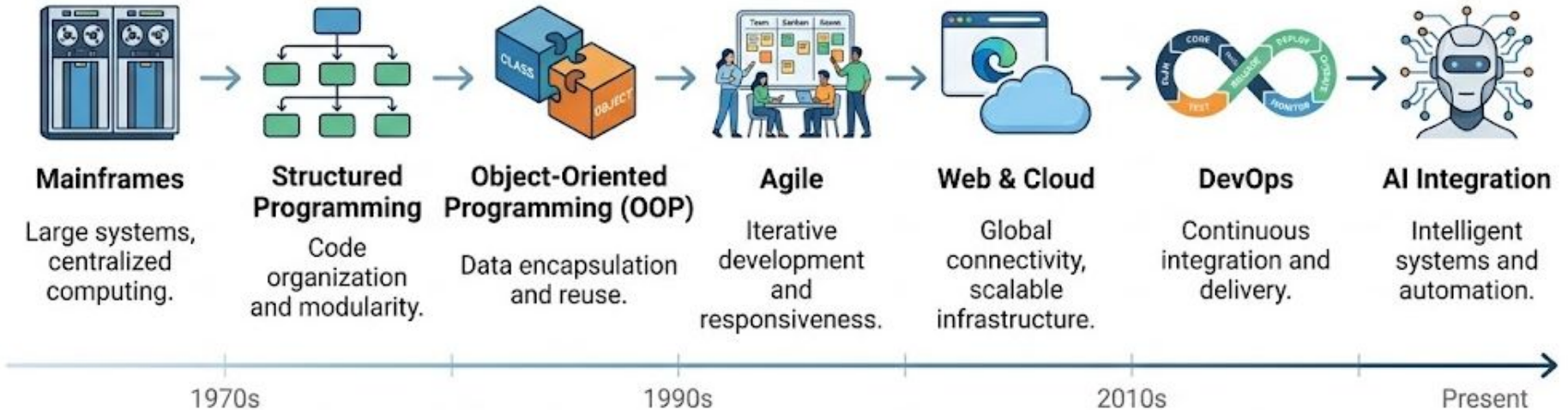
Of all the monsters that fill the nightmares of our folklore, none terrify more than werewolves, because they transform unexpectedly from the familiar into horrors. For these, one seeks bullets of silver that can magically lay them to rest.

The familiar software project, at least as seen by the nontechnical manager, has something of this character; it is usually innocent and straightforward, but is capable of becoming a monster of missed schedules, blown budgets, and flawed products. So we hear desperate cries for a silver bullet—something to make software costs drop as rapidly as computer hardware costs do.

Software has an **essential complexity** that tools alone cannot eliminate.

Fred Brooks, No Silver Bullet, 1986

Software engineering has always evolved



The tools changed. The methods changed. The role changed.

— **“Software is
eating the world,
but AI is going to
eat software”**

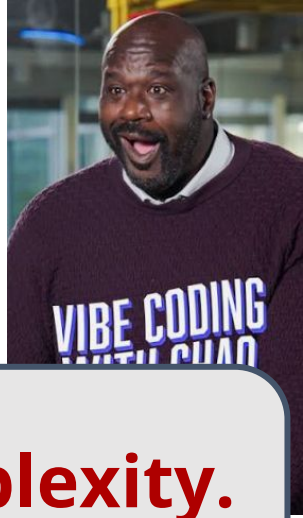
Jensen Huang in 2017



AI is changing how we build software

“This is a profound change to the nature of programming.”

Martin Fowler, Thoughtworks, May 2026



**AI can dramatically reduce accidental complexity.
Essential complexity remains.**

“You can't let the slop and cringe deny you the wonder of AI. This is the most exciting thing we've made computers do since we connected them to the internet.”

David Heinemeier Hansson, Jan 2026

Software engineering is the discipline of managing the complexity of building software

1968 NATO Conference on Software Engineering

- Provocative Title
- Call for Action
- “Software crisis”





Margaret Hamilton

- **The First 'Software Engineer'**
- "Software developers earned the right to be called engineers."
- Led the Software Engineering Division of the MIT Instrumentation Laboratory
- Contracted with NASA to develop the Apollo program's guidance system: 1961 - 1969



This Course

Our philosophy

- Conceptual background
 - often accompanied by an historical perspective
- Career preparation
- “Learning by doing”
- Develop or enhance soft skills
 - including critical thinking and problem solving, communication, teamwork and leadership, and the capacity for lifelong learning.



Laptop multitasking hinders classroom learning for both users and nearby peers

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ARTICLE INFO

ABSTRACT

“...participants who multitasked on a laptop during a lecture **scored lower** on a test compared to those who did not multitask, and participants **who were in direct view of a multitasking peer scored lower** on a test compared to those who were not. The results demonstrate that **multitasking on a laptop poses a significant distraction to both users and fellow students** and can be **detrimental** to comprehension of lecture content.”



AI and Learning

A brief detour

Article | [Open access](#) | Published: 14 April 2020

Habitual use of GPS negatively impacts spatial memory during self-guided navigation

[Louisa Dahmani](#) & [Véronique D. Bohbot](#) 

[Scientific Reports](#) 10, Article number: 6310 (2020) | [Cite this article](#)

- Extensive GPS use led to a decline in spatial memory rather than the other way around
- Poorer cognitive mapping abilities and landmark encoding

Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task[△]

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The Impact of Generative AI on Critical Thinking: Self-Reported Reductions in Cognitive Effort and Confidence Effects From a Survey of Knowledge Workers

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Thinking—Fast, Slow, and Artificial: How AI is Reshaping Human Reasoning and the Rise of Cognitive Surrender

https://doi.org/10.31234/osf.io/yk25n_v1

[The Wharton School Research Paper](#)

58 Pages • Posted: 2 Feb 2026 • Last revised: 10 Feb 2026

Steven D Shaw

University of Pennsylvania - The Wharton School; King's College London

Gideon Nave

University of Pennsylvania - The Wharton School

Date Written: January 11, 2026

How should you use AI for learning?

Discuss with the person next to you:

1. What is AI actually good for when you are learning something?
2. How do you know when AI is helping you learn, and when it is taking away an opportunity to learn?
3. How can you use AI to support your learning without giving up the parts you need to learn yourself?



Generative AI without guardrails can harm learning: Evidence from high school mathematics

Hamsa Bastani , Osbert Bastani, Alp Sungu  , , and Rei Mariman [Authors Info & Affiliations](#)

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*“... we also find that **carefully designed safeguards**, ... , can mitigate these negative effects.”*

Back to the course

AI Usage

- You may use generative AI for any and all coding tasks in the course
- We **Very Strongly Discourage** using AI to Generate writing responses, but we encourage its use for editing your writing.
- You are liable for factually inaccurate answers or unspecific rambling produced by AI tools
- It is your responsibility to validate and verify AI-produced content before submitting it for class purposes.

Course infrastructure and logistics

- Infrastructure/source of truth
 - Course website: schedule, slides, syllabus, office hours
 - Gradescope for homework, grades, other material
 - Slack for communication and collaboration.
 - Git/Github for coding and collaboration
- Logistics
 - Lecture in-person only
 - All recitations are in-person

Connect with us for the class

- All links on our course website: <https://cmu-17313q.github.io>
- We sent you an invite for Slack, please check your email.

Course Themes

- Software engineering as a human process
- Requirements and Specifications
- Metrics and Measurement
- Software Quality: Testing + CI + Security
- Continuous Deployment and DevOps
- Software Project Teams
- Managing Time, Teams, and Risks
- Software Architecture and Design Docs
- Scaling and Performance, Trade-offs
- AI/ML in Software Engineering
- Open-Source Software
- Strategic Thinking about Software

Gaining Experience: Central to 313!

- Case study analyses
- Team assignments
- Open-source engagement
- Hands-on experience is key!!!

Evaluation

- Assignments (40% projects, 5% problem sets)
 - Regular homework, mostly in teams with individual component
 - Open-source engagement
- Midterms (30 %)
- Quizzes (5%)
 - About 8-10 during the semester, schedule to be defined
- Participation activities (20 %)
 - In-class exercises
 - Pre-class reading assignments
 - Recitation exercises

Participation Activities

- Reading assignments for some lectures
 - Preparing in-class discussions: background material, case descriptions, possibly also podcast, video, Wikipedia
- In-person activities
 - **Lecture:** Active learning exercises every lecture
 - **Recitation:** Working sessions, submission on Gradescope
- All of the above count as graded “participation activities”
 - You may miss up to 4 participation activities with no grade penalty (No need to send emails ahead-of-time)

Recitations

- Practical tasks, preparation for homework, extra material, discussions
- **Have your GitHub account at the ready.**
 - Bring your laptop!
- From P2 onward: Project teams will all be in the same recitations... Good (forced) opportunity to meet in person
- We will track attendance

“Homework” Assignments / Projects *

- P1: Setup and improve a large existing software product
 - Get up-to-speed with new technologies quickly and on your own
 - P2: Collaborative development on a large software project
 - Add features and follow SE process
 - P3: Continuous Integration + Deployment
 - P4: Develop a design doc, microservices
 - P5: Open-source Excursion
 - Open-ended project: contribute to an OSS project using everything you have learned; get kudos for having PRs merged
- * subject to change

Warning! Course & HW structure may be different than what you are used to...

- Lecture topics are on high-level ideas about software engineering; case studies and experiences
- Projects require applying these ideas to technical artifacts
- Projects simulate “real-world” professional SE experience
- Technical aspects of project will not be taught in class
 - Explicit learning goal: learn new tools, languages, etc. on your own
 - Ask for help when needed; recitations provide demos and resources
- Project requirements are often **vague or under-specified** (intentionally)
 - Feel free to ask for clarifications, but expect subjective responses
 - Focus for assessment is engagement, not absolute correctness

Professionalism

- Being a professional means, you must work well with others
- The best professionals are those who make those around them better
- If you feel someone is not treating you or someone else in a professional manner, you have two options:
 - If you feel you have the standing to do so, speak up!
 - Reach out to the course staff, and we will meet with you privately to discuss it, as well as preserve your anonymity

Final Projects

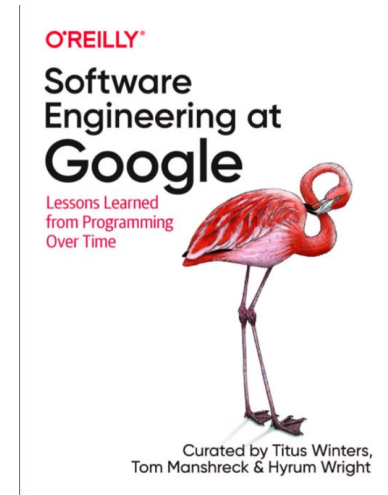
- Open-source excursion is the most fun part of the course!
- Very open-ended project. >10% of overall grade.
- Brings together everything you will have learned from lecture and prior assignments
- Teamwork and communication is very important
- In-person presentation in finals week (no exam)
- ***Do NOT book flight tickets for the winter break until finals are scheduled.***

Academic Honesty

- Standard Collaboration Policy
- In group work, be honest about contribution of group members; do not cover for others
- DO NOT submit participation sheets for people who are not in class. This is considered an academic integrity violation

Textbook

- No single textbook
- Assigned readings from different sources
 - Book chapters (library)
 - News articles
 - Lecture notes
- Recommended supplementary reading: Software Engineering at Google
 - Available for free online (legally!):
<https://abseil.io/resources/swe-book>



For next class: pre-lecture reading quiz

